AI-driven Lung-RADS Classification on CT Reports

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***Abstract*—Lung cancer has the highest mortality rate among all cancer types, affecting both men and women. It is estimated that lung cancer accounts for 21% of cancer deaths in each gender worldwide. In Brazil, lung cancer is the third most common type of cancer among men and the fourth most common among women. This alarming statistic highlights the significant impact of lung cancer on overall cancer mortality, underscoring the urgent need for effective prevention, early detection, and treatment strategies to combat this disease. The Lung-RADS is a standardized classification system for lung nodules detected in imaging exams and assesses the risk of malignancy (cancer) in these nodules to guide subsequent management decisions. In this context, the main goal of this work was to evaluate the effectiveness of using question-answering natural language processing using LLMs to extract lung nodule characteristics from Portuguese chest CT reports for automated Lung-RADS classification. Our most effective model was Llama 3.3 70B, and this model achieved a weighted F1 score of 84.77%. Our findings underscore the potential of LLMs to support radiologists in accurately categorizing lung nodules according to Lung-RADS criteria, thereby simplifying the diagnostic process.**

***Index Terms*—Lung Cancer, Lung-RADS, Natural Language Processing (NLP), Information Extraction, Large Language Models (LLMs).**

I. Introduction

Lung cancer ranks as the second most frequently diagnosed cancer among both men and women, following only breast cancer in prevalence. Despite advancements in early detection and treatment, it remains the leading cause of cancer-related mortality, responsible for more deaths in 2020 than breast and prostate cancers combined [21]. In Brazil, lung cancer is the third most common type of cancer among men and the fourth most common among women. Estimates from 2018 show that there were 18,740 new cases of lung cancer in men and 12,530 in women [14]. This statistic emphasizes the substantial contribution of lung cancer to overall cancer mortality, highlighting the need for robust strategies in prevention, early detection, and treatment to address this disease effectively [21].

Determining follow-up examinations based on the Lung-RADS guidelines for lung cancer screening CT scans is a relatively straightforward process for individuals enrolled in a lung cancer screening program. However, extracting relevant clinical information from radiology reports and organizing it according to the Lung-RADS criteria presents significant challenges. Besides, because clinical data are recorded in free-text format, this task can be complicated by the nuances of languages, such as Portuguese, which may include acronyms, negation adverbs, and complex grammatical errors that can hinder accurate interpretation. Furthermore, cultural differences and variations in descriptive style can lead to inconsistencies in the data. Finally, human error in data entry is also possible [4].

Natural language processing (NLP) is a branch of artificial intelligence that enables machines to understand, interpret, and generate human language. In the healthcare sector, NLP has greatly improved our ability to manage and analyze large volumes of textual data, such as medical records and clinical notes [17]. One important task in NLP is question answering (QA), which focuses on developing systems that automatically answer user queries in natural language by understanding the intent, retrieving relevant information, and formulating a coherent response [10]. Previous information extraction (IE) research has primarily concentrated on Chinese and English idioms to identify clinical entities, lung nodules, tumors, and associated characteristics, leaving a gap in understanding this phenomenon in other languages. Additionally, the lack of publicly available datasets containing pulmonary nodule reports hinders the development and evaluation of IE methods, especially for radiology reports in multiple languages [11] [12].

Deep learning has made significant strides in medical NLP, addressing rule-based limitations in handling complex medical terminology [12]. Convolutional Neural Networks (CNNs) have been applied to identify medical entities in records [4], though with varying results. Transformer-based models, such as BERT, have become state-of-the-art, capturing complex textual relationships [6]. Domain-specific models like BioBERT, ClinicalBERT, and PubMedBERT [13] further enhance performance by pre-training on medical corpora. Large language models (LLMs) like GPT-4, LLaMA, Gemini, and PaLM have demonstrated superior capabilities across various NLP tasks [15]. The increasing interest in applying medical LLMs in healthcare research highlights their potential to improve efficiency, accuracy, and patient care [2].

In this context, the main goal of this work was to evaluate the effectiveness of using question-answering natural language processing using Large Language Models to extract lung nodule characteristics from Portuguese chest CT reports for automated Lung-RADS classification.

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II. Material and Methods

Figure 1 details the methodology used for a QA task for Lung-RADS classification, focusing on extracting structured data from CT scan reports. The process begins with the original CT report. This report is combined with a specific prompt, a question designed to guide the LLM in extracting relevant information about lung nodules to Lung-RADS classification. The prompt can be zero-shot CoT or few-shot. The LLM processes the combined report and prompt, generating a free text response. This response is then converted through post-processing into a structured format, typically a JSON table. This table contains key-value pairs representing specific Lung-RADS criteria (e.g., "Does the nodule have solid or soft tissue attenuation?") and their corresponding values (e.g., "Yes"). The final output is a structured response, and the JSON table is ready for analysis and use. The structured data are then evaluated against a gold standard to evaluate the effectiveness of LLMs for the Lung-RADS classification.

A. Lung-RADS

The Lung Imaging Reporting and Data System, established by the American College of Radiology, serves as a standardized reporting framework to optimize the assessment and management of pulmonary nodules identified via low-dose computed tomography screenings. In the Lung-RADS classification, Category 0 refers to incomplete evaluations, where the lungs cannot be fully assessed, requiring additional chest CT scans. Category 1 indicates a negative screening result, with no pulmonary nodules identified, and requires a 12-month follow-up. Benign nodules, determined by specific features found in the imaging exam, are categorized as Category 2, with subcategories that are specified according to the size and characteristics of the pulmonary nodules. Category 3 represents probably benign lesions, requiring a 6-month follow-up for stability assessment. Categories 4A and 4B are associated with suspicious and highly suspicious nodules, respectively, each with specific follow-up protocols, including shorter intervals (3 months) for the CT scan and considering additional imaging modalities such as PET/CT. Category 4X encompasses nodules with additional features that increase the suspicion of lung cancer. Lung-RADS also incorporates an "S" modifier to accommodate clinically significant findings unrelated to lung cancer [1].

B. Dataset Annotation

The 963 chest high-dose CT reports in Portuguese were collected from January 1, 2022, to April 3, 2023, at the University Hospital of Alagoas. The research was approved by the ethics and research committee of the Federal University of Alagoas with the number: 74747817.4.0000.5013. After obtaining patient consent, all chest CT reports, irrespective of the clinical indication, were included. It is important to highlight that all patient data was anonymized. Data cleaning was applied to all reports, which included removing special characters, adding spaces between words, and removing emojis in the text. From our database of 963 reports, only those that contained answers to all questions in Table I, and only one lung nodule described in the text were selected. After the filtering process, the dataset was narrowed down to 323 reports. We split the 323 reports into 70% (226) for examples to document similarity, while the remaining 30% (97) were used as input examples for zero-shot and few-shot learning.

Based on the Lung-RADS guidelines [1], the authors defined six questions related to pulmonary nodules for the QA task. A thoracic radiologist with 15+ years of experience provided the answers to the questions used for QA for the purpose of generating a supervised dataset. Table I presents the questions and their corresponding statistics. The analysis of the 226 reports used to perform the QA procedure identified pulmonary nodules in all reports, with Lung-RADS categorization as follows: Category 0 (32 cases), Category 1 (132 cases), Category 2 (39 cases), Category 3 (7 cases), Category 4A (6 cases), Category 4B (0 cases), and Category 4X (6 cases). The analysis of the 97 reports used to perform the QA procedure identified pulmonary nodules in all reports, with Lung-RADS categorization as follows: Category 0 (14 cases), Category 1 (57 cases), Category 2 (16 cases), Category 3 (3 cases), Category 4A (4 cases), Category 4B (0 cases), and Category 4X (3 cases).

C. Models for Question Answering

For the LLMs models, we used the specific versions: GPT-4o (gpt-4o-mini-2024-07-18) [16], Gemini 2.0 Flash (gemini-2.0-flash) [22], Llama 3.3 70B [9] [15] , and DeepSeek-V3 (deep-seek-chat) [5]. In the context of LLMs, the temperature parameter regulates the uncertainty or randomness in the generation process. This parameter typically ranges from 0 (completely deterministic) to 1 or higher (resulting in increasingly random and diverse outputs) [12]. For the GPT-4o, Gemini 1.5 Flash, Llama 3.3 70B, and DeepSeek models, the temperature was set to 0 to minimize randomness in response generation. Using a lower temperature, we limited the model's tendency to take creative leaps, ensuring more predictable and consistent outputs. This is important in information extraction (IE) tasks, where the accuracy of the information extracted is crucial. Requests for the three LLM models were made via the API. We used the OpenAI API for the GPT-4o and DeepSeek-V3 models, the Together AI API for Llama 3.3 70B, and the Google AI API for the Gemini 2.0 Flash model.

To evaluate the consistency of the responses of these LLMs, the approach of [11] [20] was used. This approach consists of repeatedly presenting the same query to the models and observing variations in their responses. The same query was presented three times for each LLM, and the percentage of consistency and inconsistency was calculated for each question in the Question table. We compared the variations in the responses obtained in the three requests made to the LLMs.

The consistency percentage is given by:

$$C_p = \frac{n_c}{n_t} \cdot 100 \tag{1}$$

The inconsistency percentage is given by:

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TABLE I: Pulmonary nodules Questions and the statistics of the annotated answers.

No.	Question	Answer type	Answer statistic
1	Does the nodule have solid or soft tissue attenuation?	Boolean	20 (Positive)
2	Does the nodule have ground-glass attenuation?	Boolean	2 (Positive)
3	Does the nodule have a spiculated, irregular, or ill-defined border?	Boolean	3 (Positive)
4	Is the nodule calcified	Boolean	57 (Positive)
5	Nodule location	Categorical	25 (RUL)
			13 (RML)
			24 (RLL)
			15 (LUL)
			11 (LLL)
			8 (Others)
6	Nodule size	Numerical	4.93 mm SD 2.53

RUL - Right upper lobe RML - Right middle lobe RLL - Right lower lobe LUL - Left upper lobe LLL - Left lower lobe

$$C_n = \frac{n_{ic}}{nt} \cdot 100 \tag{2}$$

were, C_p is the consistency percentage, C_n is the inconsistency percentage, n_c is the total number of consistent responses, n_ic is the total number of inconsistent responses, and n_t is the total number of questions in the data.

D. Prompt Engineering

We designed the prompts used for QA by employing zero-shot learning based on the work of Danqing Hu et al. [11]. Besides, zero-shot chain-of-thought (CoT) [23], and few-shot learning were used as prompt techniques. The prompt templates consist of three parts: (1) Original CT report; (2) IE instructions and an unfilled Question table; and (3) Additional requirements for the IE task. In this work, the LLMs were instructed to respond with "No" as the default answer for questions that do not have corresponding information in the given CT report. Annotated reports with completed tables were provided to improve LLM task comprehension.

To effectively calculate document similarity, especially when dealing with textual data such as CT reports, we used word embeddings to transform documents into dense vectors that capture their contextual meaning. This allows us to represent documents as numerical vectors in a high-dimensional space, where semantically similar documents are positioned closer together. We used the (paraphrase-multilingual-mpnet-base-v2) model [19], available in the Hugging Face library, to calculate the embeddings of the training and test reports. Next, the cosine similarity was then used to calculate the similarity of the test reports with the training reports. This step facilitated the identification of similar training examples for few-shot learning.

The testing framework used one prompt for zero-shot learning and one for few-shot learning. For few-shot learning, it leveraged one prompt template instantiated with five examples. CT reports were combined with prompt templates to generate answers to questions. This combined prompt is submitted via API to the LLMs, and their responses are obtained. A new request is made for each CT report, preventing previous requests from influencing the IE results. Besides, the LLMs' responses are requested in JSON format to facilitate post-processing of the results. The responses from these language models do not always consist solely of the completed Question table. Therefore, any additional text is disregarded, as it is irrelevant to the analysis. The focus is exclusively on extracting the content in the form of the Question table.

E. Post-processing for Question Answering

The LLMs were instructed to extract only answers from the table provided in the prompts, but the responses were sometimes not structured. To address this issue, post-processing was applied to convert unstructured responses into a structured format. This post-processing involved removing all text except the Questions and Answers table. Moreover, when the LLM did not provide an answer in the table, leaving the space blank, or responded with "Not informed", the answer was considered "No". For Questions 1 to 4 (Table I), the answers "Yes" and "No" were converted into Boolean values. Regular expressions were used to identify the number and unit of measurement to extract the nodule size, which was then converted to millimeters. For the location Question, keywords such as "right", "left", "middle", "upper", and "lower" were employed to categorize the answer into six formats: "right upper lobe", "right middle lobe", "left lower lobe", "left upper lobe", "left lower lobe", and "others". The answer "others" was used when the specific location of the pulmonary nodule was unclear.

III. Evaluation

To evaluate the effectiveness of LLMs in the QA task using zero-shot and few-shot learning with the groudtruth dataset annotated by a thoracic radiologist, precision (P), recall (R), and F1 score (F1) were used. These same metrics were also employed to evaluate the effectiveness of combining the results of these models with a rule-based algorithm for Lung-RADS classification.

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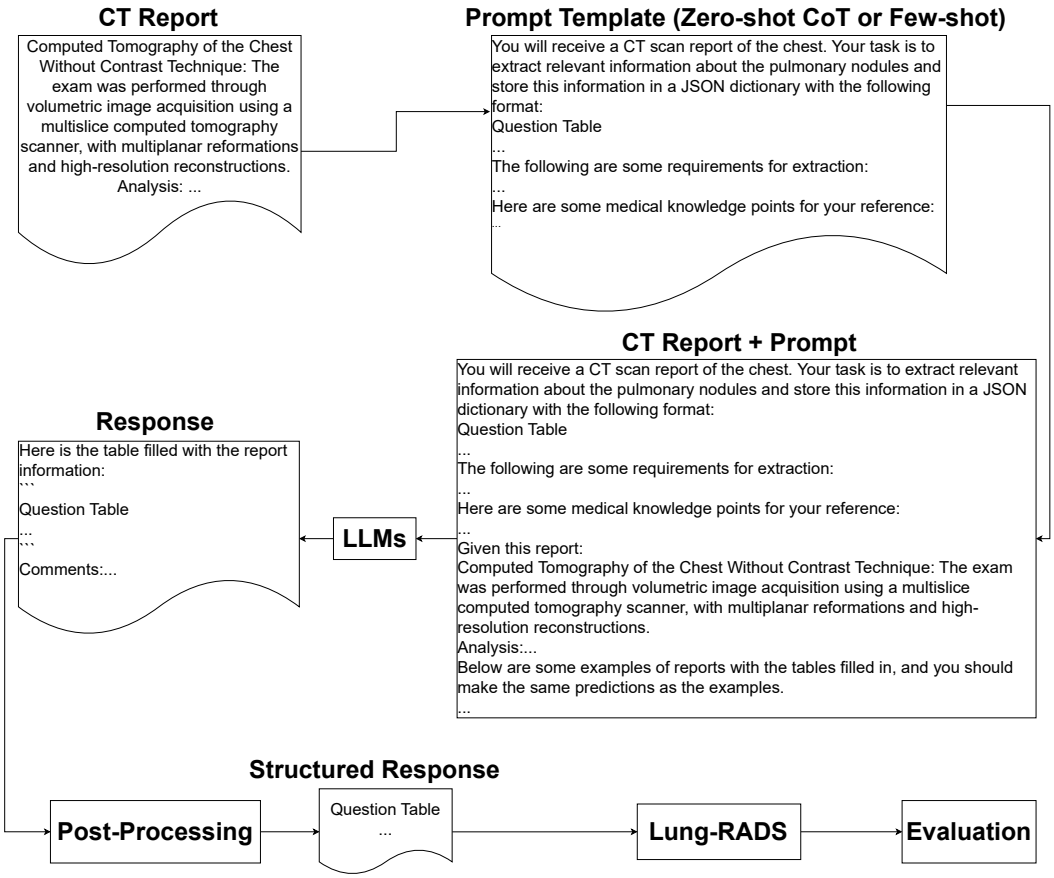

Fig. 1: Methodology scheme applied in QA task for Lung-RADS classification.

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{5}$$

where TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, respectively.

IV. Results and Discussion

The proportion of consistent responses concerning lung nodule questions for DeepSeek-V3, Gemini 2.0 Flash, GPT-4o, and Llama 3.3 70B using a zero-shot CoT template is detailed in Table II. The extraction of nodule attenuation, calcification, edges, location, and size showed greater consistency across the six prompt templates for Gemini 2.0 Flash and GPT-4o. In contrast, DeepSeek-V3 showed a reduced consistency percentage for Question 1 (lung nodule attenuation) at 86.60%, and Question 6 (lung nodule location) at 78.35%. The Llama 3.3 70B also shows a reduced consistency percentage for Question 6 with 74.23%. A potential cause for the observed behavior is that the DeepSeek-V3 and Llama 3 70B models using a zero-shot CoT prompt are extracting size and attenuation characteristics from findings not related to the lung nodules described within the chest CT report. In contrast, as shown in Table II, a few-shot learning template improves the consistency of the response. Four models achieved over 96% consistency of the response in six questions, suggesting that few-shot learning enables LLMs to generate similar responses to identical queries.

Table III shows the precision, recall, and F1 score metrics for DeepSeek-V3, Gemini 2.0 Flash, GPT-4o, and Llama 3.3 70B in the QA task using a zero-shot prompt template with CoT. DeepSeek-V3 demonstrated significantly lower F1 scores for Q1, Q2, and Q3 questions. The specific errors that contributed to these low scores were as follows:

- Q1: In 29 cases where pulmonary nodules were described as having calcification, the DeepSeek-V3 model incorrectly identified "solid attenuation" as a characteristic, although this characteristic was not present in the reported descriptions.
- Q2: When reports included terms such as "focal areas with ground-glass attenuation", "opacities with ground-

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TABLE II: Evaluation Consistency of Response LLMs Zero-shot CoT and Few-shot Prompt Templates

LLM	Prompt	Q1	Q2	Q3	Q4	Q5	Q6	Avg Consistency
DeepSeek-V3	Zero-shot CoT	86.60	96.91	98.97	93.81	78.35	94.84	91.58
Gemini 2.0 Flash	Zero-shot CoT	100.00	98.97	100.00	100.00	100.00	95.88	99.14
GPT-4o	Zero-shot CoT	95.88	95.88	96.91	96.91	91.75	94.84	95.36
Llama 3.3 70B	Zero-shot CoT	92.78	96.91	98.97	100.00	74.23	96.91	93.29
DeepSeek-V3	Few-shot	100.00	100.00	100.00	100.00	96.91	98.97	99.31
Gemini 2.0 Flash	Few-shot	98.97	100.00	98.97	100.00	98.97	98.97	99.28
GPT-4o	Few-shot	98.97	100.00	100.00	97.94	97.94	96.91	98.63
Llama 3.3 70B	Few-shot	97.99	100.00	100.00	98.97	97.94	97.94	98.81

glass attenuation" or "similar", the model incorrectly assigned "ground-glass attenuation" to the target pulmonary nodule, even when this feature was associated with other distinct areas within the report.

- Q3: Similarly, in reports containing phrases such as "focal, ill-defined centrilobular opacities" or "poorly defined hypoattenuating parenchymal images", DeepSeek-V3 incorrectly extracted "ill-defined borders" as a characteristic of the target pulmonary nodule, again misinterpreting the characteristics association with other described areas.

Gemini 2.0 Flash model shows a lower F1 score for Question 1. This was primarily due to the following error:

- Q1: When reports included terms such as "noncalcified nodule", "noncalcified opacity", or "noncalcified nodular formation", the model incorrectly extracted the "solid attenuation" characteristic, which was not a characteristic of the described pulmonary nodule.

GPT-4o model demonstrated lower F1 scores for Q2 and Q3, mainly due to misattribution of characteristics:

- Q2: When reports mentioned "opacities with ground-glass attenuation", "focal area with ground-glass attenuation", "ground-glass micronodule" or "ground-glass lesions", the model incorrectly assigned "ground-glass attenuation" to the target pulmonary nodule, despite this feature related to other findings.
- Q3: For reports that contained terms such as "poorly defined centrilobular opacities", "poorly defined hypoattenuating parenchymal images", the model extracted the characteristic "poorly defined edges", this characteristic being related to other findings in the report and not belonging to the pulmonary nodule analyzed.

Llama 3.3 70B exhibited lower F1 scores for Q2 due to misattribution of characteristics:

- Q2: When reports mentioned "opacities with ground-glass attenuation" or "irregular reticular opacities", the model incorrectly assigned "ground-glass attenuation" to the target lung nodule, despite this feature related to other findings.

Table IV details the precision, recall, and F1 score metrics for DeepSeek-V3, Gemini 2.0 Flash, GPT-4o, and Llama 3.3 70B in a QA task using a few-shot prompt template with five provided examples. DeepSeek-V3 demonstrated an increase in F1 scores for Questions 1 and 2. Gemini 2.0 Flash exhibited an increase in the F1 score for Question 1 while experiencing a decrease for Question 2. GPT-4o's F1 scores improved for Questions 1 and 2 but saw a slight reduction for Questions 3 and 4. Llama 3.3 70B showed higher F1 scores on Questions 1, 2, and 3. A potential explanation for these variations in F1 scores is the inclusion of input examples, which likely contributes to the LLMs becoming more accurate in characteristics extraction. The questions that resulted in low F1 scores for all four models exhibited the same error behaviors as those previously detailed for the models using the zero-shot CoT prompt.

Figure 2a shows a comparision of four LLMs (DeepSeek-V3, Gemini 2.0 Flash, GPT-4o, and Llama 3.3 70B) across six questions using the F1 score. The results suggest that the Gemini 2.0 Flash and Llama 3.3 70B have a higher F1 score across the different questions. DeepSeek-V3 and GPT-4o show lower overall effectiveness and greater variability, particularly struggling with Question 2. The Friedman test [18] was conducted to identify statistical differences between the four LLMs. After that, the Nemenyi [18] test was done to determine which ones are significantly different. The Friedman test yielded a $p - value = 0.0127$, indicating a statistically significant difference between at least two models, but this test does not provide the models that are statistically different. However, the Nemenyi post-hoc test did not detect statistically significant differences ($p - value < 0.05$) between specific model pairs. This suggests that while there is an overall difference, the pairwise differences may not be strong enough to be detected by the Nemenyi test.

During the tests conducted in this study, the cost of GPT-4o was US$0.075 per 1 million tokens for input and US$0.60 per 1 million tokens for output. For DeepSeek-V3, the cost was approximately US$0.07 per 1 million tokens for input and US$1.10 per 1 million tokens for output. Llama 3.3 70B and Gemini 2.0 Flash models have no cost to process incoming and outgoing tokens. Depending on the volume of text processed, the cost of these paid models may hinder their use in real-world systems. Given that there is no significant difference between the F1 scores of the four models, the absence of processing costs for Llama 3.3 70B and Gemini 2.0 Flash makes them attractive alternatives for applications where cost efficiency is a priority.

Due to the increased consistency in the responses observed

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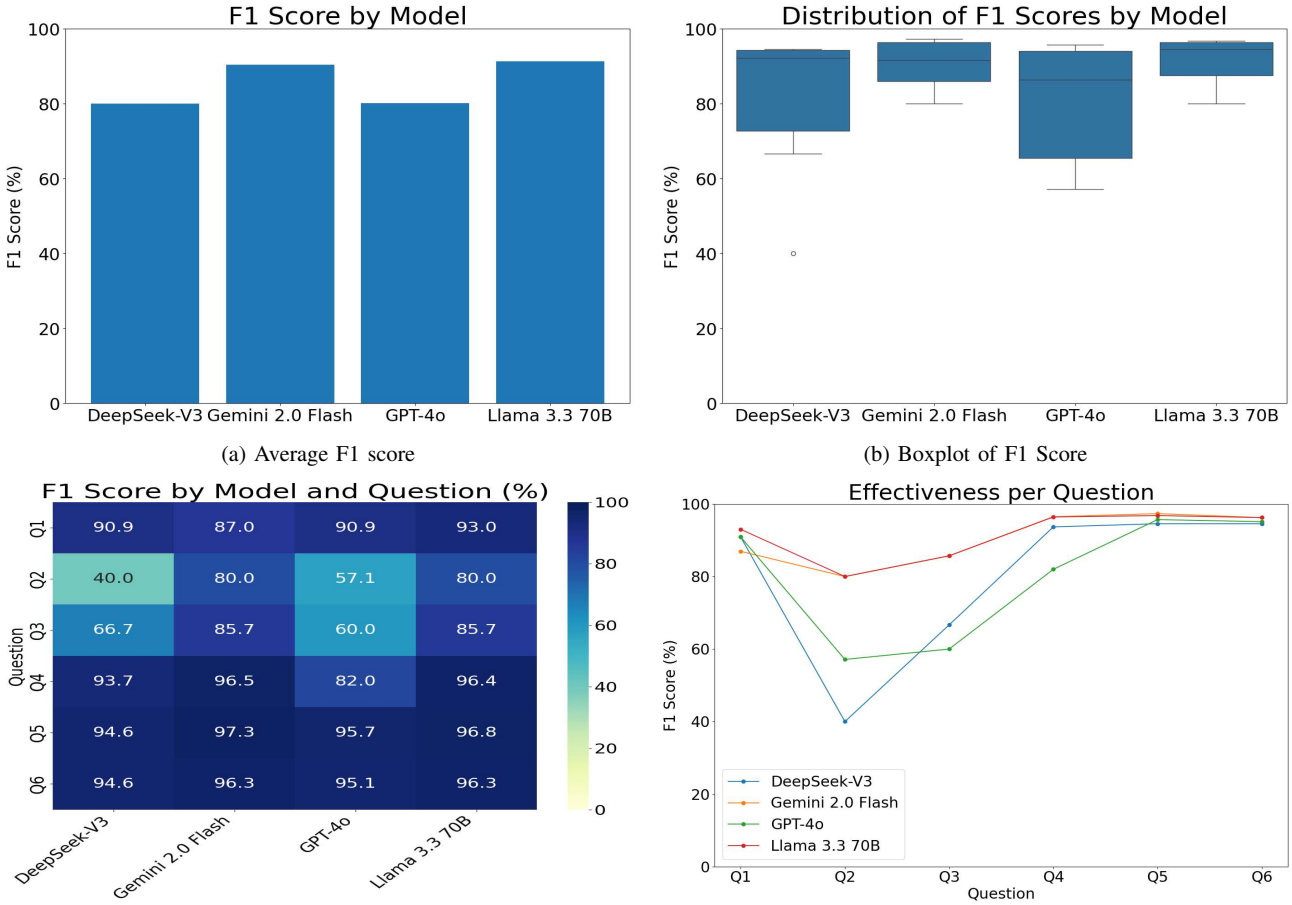
TABLE III: LLMs Effectiveness using Zero-shot CoT Prompt

Question	DeepSeek-V3			Gemini 2.0 Flash			GPT-4o			Llama 3.3 70B		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Q1	33.33	100.00	50.00	52.63	100.00	68.97	79.17	95.00	86.36	72.00	90.00	80.00
Q2	10.53	100.00	19.05	100.00	100.00	100.00	12.50	100.00	22.22	50.00	100.00	66.67
Q3	50.00	100.00	66.67	75.99	100.00	85.71	50.00	100.00	66.67	60.00	100.00	75.00
Q4	96.08	84.48	89.91	98.21	94.83	96.49	96.00	82.76	88.89	98.11	89.66	93.69
Q5	90.72	100.00	95.13	94.85	100.00	97.35	90.72	100.00	95.14	89.69	100.00	94.57
Q6	90.72	100.00	95.13	89.69	100.00	94.27	87.63	100.00	93.41	89.69	100.00	94.57

TABLE IV: LLMs Effectiveness using Few-shot with 5 Examples

Question	DeepSeek-V3			Gemini 2.0 Flash			GPT-4o			Llama 3.3 70B		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Q1	83.33	100.00	90.91	76.92	100.00	86.96	83.33	95.00	90.91	86.96	100.00	93.02
Q2	25.00	100.00	40.00	66.67	100.00	80.00	40.00	100.00	57.14	66.67	100.00	80.00
Q3	50.00	100.00	66.67	75.99	100.00	85.71	42.86	100.00	60.00	75.00	100.00	85.71
Q4	98.11	89.66	93.69	98.21	94.83	96.49	97.62	70.69	82.00	100.00	93.10	96.43
Q5	89.69	100.00	94.57	94.85	100.00	97.35	91.75	100.00	95.70	93.81	100.00	96.81
Q6	89.69	100.00	94.57	92.78	100.00	96.26	90.72	100.00	95.13	92.78	100.00	96.26



(a) Average F1 score

(b) Boxplot of F1 Score

(c) Heatmap F1 score

(d) Effectiveness per question

Fig. 2: LLMs results for the QA task with few-shot prompt

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TABLE V: Lung-RADS Classification Result Few-Shot Prompt

Lung-RADS Category	DeepSeek-V3			Gemini 2.0 Flash			GPT-4o			Llama 3.3 70B			Nº Ex.
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
0	71.43	71.43	71.43	76.92	71.43	74.07	39.29	78.57	52.38	71.43	71.43	71.43	14
1	96.23	89.47	92.73	96.43	94.74	95.58	95.24	70.18	80.81	96.23	89.47	92.73	57
2	71.43	93.75	81.08	87.50	87.50	87.50	78.95	93.75	85.71	71.43	93.75	81.08	16
3	100.00	66.67	80.00	50.00	66.67	57.14	100.00	66.67	80.00	100.00	66.67	80.00	3
4A	66.67	50.00	57.14	60.00	75.00	66.67	100.00	25.00	40.00	66.67	50.00	57.14	4
4X	50.00	66.67	57.14	75.00	66.67	66.67	40.00	66.67	50.00	50.00	66.67	57.14	3
weighted avg	86.03	84.54	84.77	88.28	87.63	87.87	83.11	73.20	74.85	86.02	84.54	84.77	97

in the four models, we utilized the results obtained from the few-shot learning strategy with five examples for Lung-RADS classification. Table V shows the precision, recall, and F1 score for DeepSeek-V3, Gemini 2.0 Flash, GTP-4o, and Llama 3.3 70B for the Lung-RADS classification. For the Lung-RADS 0 classification, the DeepSeek-V3, Gemini 2.0 Flash, and Llama 3.3 70B models showed good effectiveness (around 72%). However, the GPT-4 model obtained the lowest effectiveness (52.38%), probably because it could not extract the "calcification characteristic" of the nodules, leading to an incorrect classification as Lung-RADS 0. All models demonstrated high effectiveness in classifying Lung-RADS 1 nodules. This likely stems from the prevalence of this classification in the training data and the clear textual descriptions of these nodules in the reports. Besides, the absence of similar characteristics, such as "calcification", which are associated with other findings, may have contributed to this effectiveness. In the Lung-RADS 2 classification, all four models demonstrated good effectiveness, achieving approximately 85%. Gemini 2.0 Flash exhibited the highest F1 score at 87.50%, indicating the models' ability to effectively extract the distinguishing characteristics of lung nodules for this classification.

For the Lung-RADS 3 classification, DeepSeek-V3, GPT-4, and Llama 3.3 70B demonstrated good effectiveness, achieving an F1 score of 80%. However, Gemini 2.0 Flash exhibited the lowest F1 score at 57.14%. This is likely attributed to the model's inaccurate extraction of nodule calcification and attenuation characteristics. The limited number of test samples (only 3) may have also contributed to this outcome. Classifications Lung-RADS 4A and Lung-RADS 4X demonstrated the lowest effectiveness across the six Lung-RADS categories. The reduced number of test samples (four for 4A and three for 4X) within these classifications may have compromised the models' effectiveness. With such limited data, the models may have incorrectly extracted nodule characteristics such as attenuation or edge characteristics, leading to inaccurate Lung-RADS classifications.

Therefore, the results suggest that all four LLMs effectively classified lung nodules in Portuguese chest CT reports to assist radiologists with Lung-RADS indexing. The findings highlight a pathway for developing a more adaptable NLP system leveraging LLMs, which has significant implications for Lung-RADS classification tasks.

Our study wasn't the first that utilizes NLP to assign the Lung-RADS index from CT reports. Beyer et al. [3] developed a rule-based algorithm to assign the Lung-RADS classification to lung nodules identified in structured clinical reports of CT lung screening examinations in English. Their NLP algorithm achieved a sensitivity of 75% and a specificity of 98% to identify Lung-RADS 3 and 4 nodules. Unlike Beyer et al., who used a database of 1501 structured reports, we employed 963 unstructured reports; while they used a rule-based algorithm, we used LLMs for characteristics extraction via a QA task.

Gandomi et al. [7] further advanced NLP in Lung-RADS index extraction by developing a rule-based algorithm and comparing it with the performance of radiologists and Lung Cancer Screening (LCS) specialists. Their approach demonstrated high precision (99%) and recall (99%) across four ground truth sets of CT reports. Our approach differs in several ways: Gandomi et al. used a larger database of 24,060 reports, while our study utilized 963 reports, which may limit generalizability. Additionally, Gandomi's reports already included the Lung-RADS index, whereas our research assigns the Lung-RADS index based on the characteristics described within the reports.

García-Barragán et al. [8] used LLMs to extract disease phenotypic entities for disease understanding. The authors compared the effectiveness of the baseline models, including scispaCy, BioFalcon, MetaMap, and GliNER, with Llama-2 7B, Mistral 7B, GPT-3.5, and GPT-4. Their results demonstrated that GPT-4 achieved the highest F1 score (78%) among all evaluated models across the tested prompts. Our approach differs in several ways: They used a dataset derived from Wikipedia, while we used chest CT reports. They analyzed 225 documents containing disease descriptions, whereas we utilized 963 chest CT reports. Furthermore, García-Barragán et al. compared LLMs with a wide range of parameter sizes, while we focused on comparing LLMs of similar sizes.

While promising, our research is subject to several key limitations. First, the analysis relied on annotations validated by a single radiologist, which could introduce bias due to the lack of diverse medical perspectives. Second, data were sourced from a single hospital, limiting the study's generalizability to wider patient demographics and healthcare settings. Third, the test dataset for Lung-RADS classification was heavily imbalanced, skewing the results and potentially underrepresenting clinically important but less frequent categories.

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V. Conclusion

This study used LLMs to extract lung nodule characteristics from chest CT reports written in Portuguese to automate the Lung-RADS classification. We found that all four evaluated LLMs (DeepSeek-V3, Gemini 2.0 Flash, GPT-4o, and Llama 3.3 70B) effectively classified lung nodules in Portuguese chest CT reports to assist radiologists with Lung-RADS indexing. Our findings highlight a pathway for developing a more adaptable NLP system by leveraging deep learning and LLMs models, which has significant implications for Lung-RADS classification tasks.

Our findings underscore the potential of LLMs to support radiologists in accurately categorizing lung nodules according to Lung-RADS criteria, thereby simplifying the diagnostic process. By automating and improving information extraction, these models are expected to reduce radiologists' workload and enhance the consistency of follow-up recommendations.

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